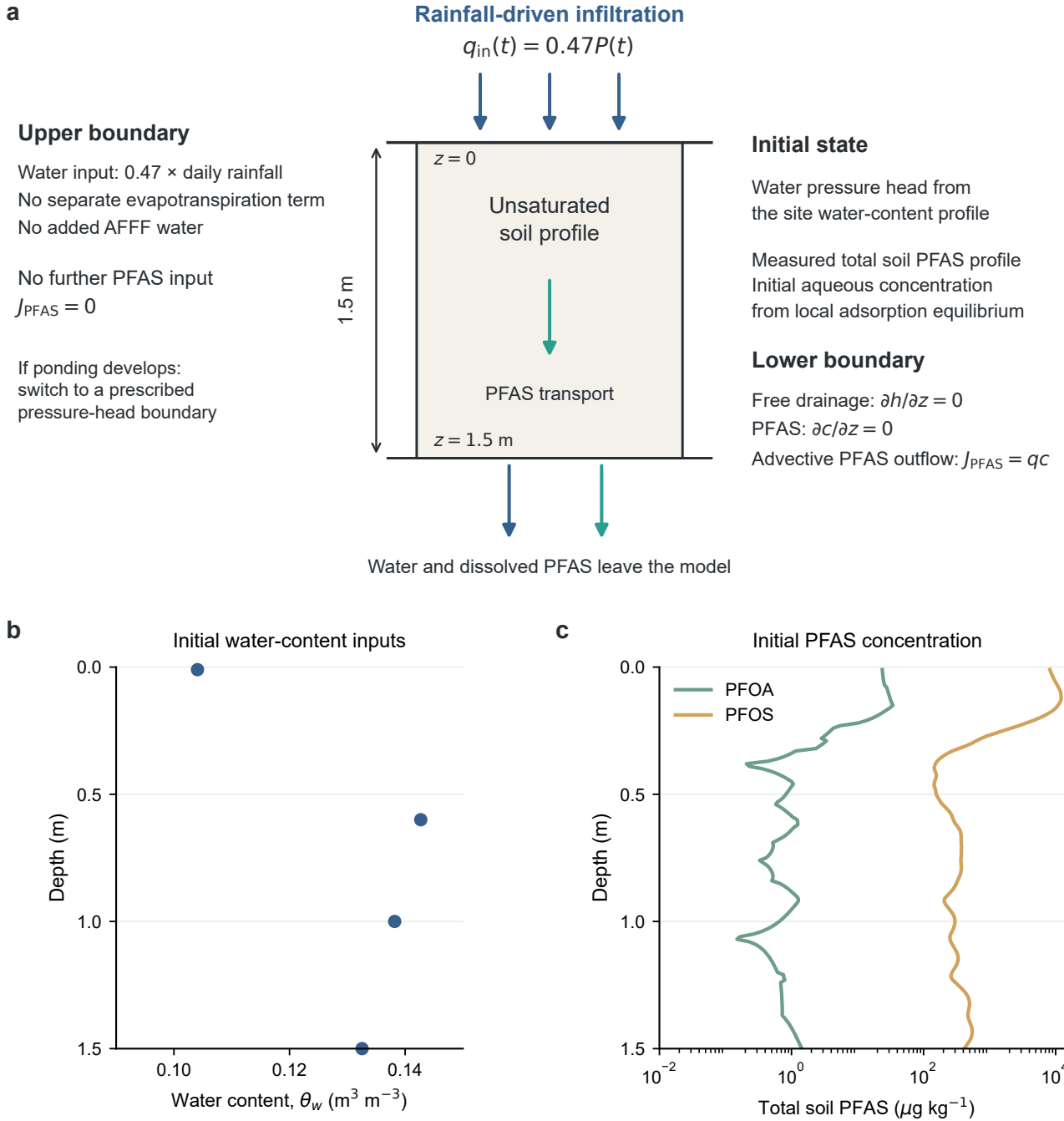

a
Rainfall-driven infiltration
$q_{in}(t) = 0.47P(t)$
Upper boundary
Water input: 0.47 × daily rainfall
No separate evapotranspiration term
No added AFFF water
No further PFAS input
$J_{PFAS} = 0$
If ponding develops:
switch to a prescribed
pressure-head boundary
z = 0
Unsaturated
soil profile
PFAS transport
z = 1.5 m
1.5 m
Initial state
Water pressure head from
the site water-content profile
Measured total soil PFAS profile
Initial aqueous concentration
from local adsorption equilibrium
Lower boundary
Free drainage: $\partial h/\partial z = 0$
PFAS: $\partial c/\partial z = 0$
Advective PFAS outflow: $J_{PFAS} = qc$
Water and dissolved PFAS leave the model
b
Initial water-content inputs
Depth (m)
0.0
0.5
1.0
1.5
0.10
0.12
0.14
Water content, θ_w (m^3 m^{-3})
c
Initial PFAS concentration
PFOA
PFOS
Depth (m)
10^{-2}
10^{0}
10^{2}
10^{4}
Total soil PFAS (μg kg^{-1})